Category Theory

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1 Categories

1.1 Categories and Morphisms

Definition 1.1 (Category)

A **category** $\mathcal{C}$ consists of two forms of data, called **objects**—denoted as $X, Y, Z, \ldots$—and **morphisms**—denoted as $f, g, h \ldots$—each of which have a domain and a codomain.[a] It satisfies the following properties.

1. *Composition.* For two morphisms $f : X \to Y$, $g : Y \to Z$, there exists a composite morphism $g \circ f : X \to Z$.
2. *Identity.* Each object has the identity morphism $1_X : XtoX$ satisfying that for all $f : X \to Y$,
$$f \circ 1_X = f = 1_Y \circ f \tag{1}$$
3. *Associativity.* For all $f : X \to Y, g : Y \to Z, h : Z \to W$,
$$h \circ (g \circ f) = (h \circ g) \circ f \tag{2}$$

[a] For example, $f : X \to Y$ indicates that the morphism f has a domain X and codomain Y.

Example 1.1 (Basic Examples)

Here are some familiar categories.

1. **Set** is the category where the objects are sets, morphisms are functions, composition is function composition, and identity morphisms are identity maps.
2. **Group** is the category where the objects are groups, morphisms are group homomorphisms, composition is function composition, and identity morphisms are identity maps.
3. **Ab** is the category where the objects are abelian groups, morphisms are group homomorphisms, composition is function composition, and identity morphisms are identity maps.
4. **Ring** is the category where the objects are rings, morphisms are ring homomorphisms, composition is function composition, and identity morphisms are identity maps.
5. **Field** is the category where the objects are fields, morphisms are field homomorphisms, composition is function composition, and identity morphisms are identity maps.
6. **Top** is the category where the objects are topolgical spaces,morphisms are continuous functions, composition is function composition, and identity morphisms are identity maps.
7. **Vec** is the category where the objects are vector spaces over $\mathbb{F}$, morphisms are linear maps, composition is function composition, and identity morphisms are identity maps.

Example 1.2 (Finite Category)

Let $\mathcal{C}$ be the category associated with the partially ordered set $(\mathcal{P}(\{1,2,3\}), \subseteq)$. Its objects are the subsets of $\{1,2,3\}$, and
$$\text{Hom}_{\mathcal{C}}(A, B) = \begin{cases} \{\iota_{A,B}\} & A \subseteq B, \\ \varnothing & \text{otherwise.} \end{cases} \tag{3}$$
We can visualize this category using its Hasse diagram.

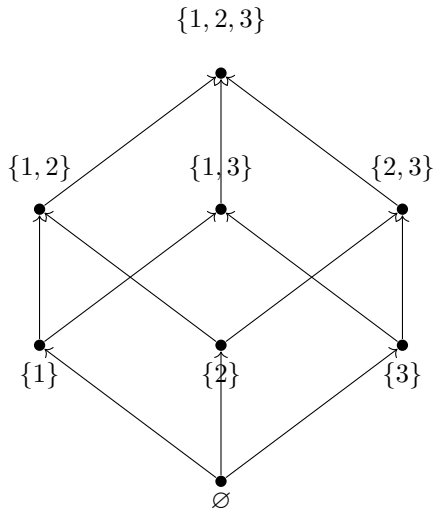


Figure 1: A finite category induced by a partial order. Identity morphisms and arrows obtained by composition are omitted.

Composition follows from the transitivity of inclusion: if $A \subseteq B \subseteq C$, then

$$\iota_{B,C} \circ \iota_{A,B} = \iota_{A,C}. \tag{4}$$

The object $\varnothing$ is initial, and $\{1, 2, 3\}$ is terminal.

Definition 1.2 (Isomorphism)

An **isomorphism** in a category $\mathcal{C}$ is a morphism $f : X \to Y$ such that there exists an inverse morphism $g : Y \to X$ satisfying

$$f \circ g = 1_Y, \quad g \circ f = 1_X \tag{5}$$

Example 1.3 (Isomorphisms)

Here are some isomorphisms in our familiar categories.

1. All bijective functions in **Set**.
2. All group isomorphisms in **Group**.
3. All ring isomorphisms in **Ring**.
4. All field isomorphisms in **Field**.
5. All homeomorphisms in **Top**.
6. All vector space isomorphisms in **Vec**.

Definition 1.3 (Notation)

Given a category $\mathcal{C}$,

1. $\text{Ob}(\mathcal{C})$ is the collection of all objects in $\mathcal{C}$.
2. $\text{Mor}(\mathcal{C})$ is the collection of all morphisms in $\mathcal{C}$.

3. $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is the collection of all morphisms $X \to Y$ in $\mathcal{C}$.
4. $\mathrm{Iso}_{\mathcal{C}}(X, Y)$ is the collection of all isomorphisms $X \to Y$ in $\mathcal{C}$.
5. $\mathrm{End}_{\mathcal{C}}(X, Y)$ is the collection of all morphisms $X \to X$ in $\mathcal{C}$.
6. $\mathrm{Aut}_{\mathcal{C}}(X, Y)$ is the collection of all isomorphisms $X \to X$ in $\mathcal{C}$.

If the context is clear on which category we are working on, we can drop the subscript $\mathcal{C}$.

Note that these collections, like $\mathrm{Hom}(X, Y)$, may not actually form a set. If they do, we call the category *locally small*.

Definition 1.4 (Locally Small Category)

A category $\mathcal{C}$ is **locally small** if the morphisms between any objects in the category form a set. In this case, we call $\mathrm{Hom}(X, Y)$ a *hom set*.

Definition 1.5 (Subcategory)

Let $\mathcal{C}$ be a category. A **subcategory** $\mathcal{D}$ of $\mathcal{C}$ is a category satisfying

1. $\mathrm{Ob}(\mathcal{D}) \subseteq \mathrm{Ob}(\mathcal{C})$.
2. For all $X, Y \in \mathrm{Ob}(\mathcal{D})$,
$$\mathrm{Hom}_{\mathcal{D}}(X, Y) \subseteq \mathrm{Hom}_{\mathcal{C}}(X, Y). \tag{6}$$
3. For every $X \in \mathrm{Ob}(\mathcal{D})$, the identity morphism id_X belongs to $\mathrm{Hom}_{\mathcal{D}}(X, X)$.
4. The composition of morphisms in $\mathcal{D}$ is the restriction of the composition in $\mathcal{C}$.

A subcategory $\mathcal{D}$ of $\mathcal{C}$ is called

1. **Full** if for every $X, Y \in \mathrm{Ob}(\mathcal{D})$,
$$\mathrm{Hom}_{\mathcal{D}}(X, Y) = \mathrm{Hom}_{\mathcal{C}}(X, Y). \tag{7}$$
2. **Wide** if $\mathrm{Ob}(\mathcal{D}) = \mathrm{Ob}(\mathcal{C})$.
3. **Proper** if $\mathcal{D} \neq \mathcal{C}$.

Example 1.4 (Subcategories)

The following are examples of subcategories.

1. The category **Ab** of abelian groups is a full subcategory of **Group**, since every group homomorphism between abelian groups is also a morphism in **Ab**.
2. The category **FinSet** of finite sets and functions between them is a full subcategory of **Set**.
3. The category $\mathbf{FinVect}_k$ of finite-dimensional vector spaces over a field k and linear transformations between them is a full subcategory of $\mathbf{Vect}_k$.
4. The category **Haus** of Hausdorff topological spaces and continuous maps between them is a full subcategory of **Top**.
5. The category **Top** with only homeomorphisms as morphisms is a wide subcategory of **Top**, since it has the same objects but fewer morphisms.
6. The category of sets with injective functions as morphisms is a wide subcategory of **Set**, since the composition of injective functions is injective.
7. The category of sets with bijective functions as morphisms is a wide subcategory of **Set**.
8. The category of vector spaces over k with only invertible linear transformations as morphisms is a wide subcategory of $\mathbf{Vect}_k$.

1.2 Functors

Definition 1.6 (Functor)

A **functor** $F : \mathcal{C} \to \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$ is a mapping which associates objects X and morphisms f in $\mathcal{C}$ to objects $F(X)$ and morphisms $F(f)$ in $\mathcal{D}$. It must satisfy the properties:

1. For any objects X, Y in $\mathcal{C}$, $F(f : X \to Y) = F(f) : F(X) \to F(Y)$.
2. For any morphisms f, g in $\mathcal{C}$, $F(g \circ f) = F(g) \circ F(f)$.
3. $F(1_X) = 1_{F(X)}$

$$\begin{array}{ccc} X & \longmapsto & F(X) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle F(f)} \\ Y & \longmapsto & F(Y) \end{array}$$

This allows us to build the category of categories.

Example 1.5 (Cat)

Cat is the category consisting of categories as its objects and functors as its morphisms.

Definition 1.7 (Forgetful Functor)

A functor $U : \mathcal{C} \to \mathcal{D}$ is a **forgetful functor** if it sends objects and morphisms of $\mathcal{C}$ to their underlying objects and morphisms in $\mathcal{D}$, discarding some of the additional structure present in $\mathcal{C}$.[a]

[a] Unlike the functor, there is no single universally accepted definition of what a forgetful functor is.

Example 1.6 (Forgetful Functors)

The following are examples of forgetful functors.

1. The functor $U : \mathbf{Group} \to \mathbf{Set}$ that associates each group (object in **Group**) with its underlying set (object in **Set**) and each group homomorphism (morphism in **Group**) with its underlying function (morphism in **Set**).
2. The functor $U : \mathbf{Ab} \to \mathbf{Group}$ that associates each abelian group with its underlying group and each abelian group homomorphism with the same group homomorphism, forgetting the requirement of commutativity.
3. The functor $U : \mathbf{Ring} \to \mathbf{Ab}$ that associates each ring with its underlying additive abelian group and each ring homomorphism with its underlying additive group homomorphism, forgetting multiplication.
4. The functor $U : \mathbf{Ring} \to \mathbf{Set}$ that associates each ring with its underlying set and each ring homomorphism with its underlying function, forgetting both addition and multiplication.
5. The functor $U : \mathbf{Vect}_k \to \mathbf{Ab}$ that associates each vector space over k with its underlying additive abelian group and each linear transformation with its underlying group homomorphism, forgetting scalar multiplication.
6. The functor $U : \mathbf{Vect}_k \to \mathbf{Set}$ that associates each vector space over k with its underlying set and each linear transformation with its underlying function, forgetting vector addition and scalar multiplication.
7. The functor $U : \mathbf{Top} \to \mathbf{Set}$ that associates each topological space with its underlying set and each continuous map with its underlying function, forgetting the topology.

8. The functor $U : \mathbf{Met} \to \mathbf{Top}$ that associates each metric space with its induced topological space and each continuous map with the same continuous map, forgetting the metric but retaining the induced topology.
9. The functor $U : \mathbf{Mod}_R \to \mathbf{Ab}$ that associates each left module over a ring R with its underlying additive abelian group and each module homomorphism with its underlying group homomorphism, forgetting scalar multiplication by elements of R.

Definition 1.8 (Full Functors)

A functor $F : \mathcal{C} \to \mathcal{D}$ is **full** when $F_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is surjective for any objects $X, Y \in \mathrm{Ob}(C)$.

Definition 1.9 (Faithful Functor)

A functor $F : \mathcal{C} \to \mathcal{D}$ is **faithful** when $F_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is injective for any objects $X, Y \in \mathrm{Ob}(C)$.

Definition 1.10 (Inclusion Functor)

Let $\mathcal{D}$ be a subcategory of $\mathcal{C}$. The **inclusion functor** is the functor

$$I : \mathcal{D} \hookrightarrow \mathcal{C} \tag{8}$$

defined by

1. *On objects.* For every $X \in \mathrm{Ob}(\mathcal{D})$,
$$I(X) = X. \tag{9}$$
2. *On morphisms.* For every $f \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$,
$$I(f) = f. \tag{10}$$

The inclusion functor is always faithful, and it is full if and only if $\mathcal{D}$ is a full subcategory of $\mathcal{C}$.

Proof.

Example 1.7

Consider the forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$.

1. U is faithful since for any two groups $G, H \in \mathrm{Ob}(\mathbf{Grp})$, a homomorphism $\varphi : G \to H \in \mathrm{Mor}(\mathbf{Grp})$ is mapped to a unique function $U(\varphi) \in \mathrm{Mor}(\mathbf{Set})$. Therefore, $U_{G,H} : \mathrm{Hom}_{\mathbf{Grp}}(G, H) \to \mathrm{Hom}_{\mathbf{Set}}(U(X), U(Y))$ is injective.
2. U is not full since there exists functions in $\mathrm{Mor}(\mathbf{Set})$ that are not homomorphisms.

Definition 1.11 (Opposite Category)

Let $\mathcal{C}$ be a category. The **opposite category** $\mathcal{C}^{\mathrm{op}}$ is the category obtained by reversing the direction of every morphism in $\mathcal{C}$. More precisely,

1. *Objects.* The objects are the same:
$$\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C}). \tag{11}$$
2. *Morphisms.* For every pair of objects X, Y,
$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathcal{C}}(Y, X). \tag{12}$$
3. *Composition.* If $f : X \to Y$ and $g : Y \to Z$ are morphisms in $\mathcal{C}^{\mathrm{op}}$, their composition is defined by
$$g \circ_{\mathrm{op}} f = f \circ g, \tag{13}$$
where the composition on the right is taken in $\mathcal{C}$.
4. *Identity.* The identity morphisms are unchanged:
$$1_X^{\mathrm{op}} = 1_X. \tag{14}$$

In particular,
$$(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}. \tag{15}$$

Example 1.8 (Opposite Categories)

The following are examples of opposite categories.

1. Let $\mathcal{C}$ be the category
$$X \xrightarrow{f} Y \xrightarrow{g} Z \tag{16}$$
together with identities and compositions. Then $\mathcal{C}^{\mathrm{op}}$ has the reversed arrows
$$X \xleftarrow{f^{\mathrm{op}}} Y \xleftarrow{g^{\mathrm{op}}} Z \tag{17}$$
satisfying
$$f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}. \tag{18}$$
2. Every partially ordered set $(P, \leq)$ can be regarded as a category in which there exists a unique morphism $x \to y$ if and only if $x \leq y$. Its opposite category corresponds to the reversed partial order $(P, \geq)$.
3. An object I is initial in $\mathcal{C}$ if and only if it is terminal in $\mathcal{C}^{\mathrm{op}}$. Similarly, terminal objects in $\mathcal{C}$ are initial objects in $\mathcal{C}^{\mathrm{op}}$.

Definition 1.12 (Covariant and Contravariant Functors)

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

1. A **covariant functor** $F : \mathcal{C} \to \mathcal{D}$ preserves the direction of morphisms. That is, for every morphism $f : X \to Y$ in $\mathcal{C}$,
$$F(f) : F(X) \to F(Y). \tag{19}$$
2. A **contravariant functor** $F : \mathcal{C} \to \mathcal{D}$ reverses the direction of morphisms. That is, for every morphism $f : X \to Y$ in $\mathcal{C}$,
$$F(f) : F(Y) \to F(X). \tag{20}$$
It satisfies
$$F(1_X) = 1_{F(X)}, \qquad F(g \circ f) = F(f) \circ F(g) \tag{21}$$
for every pair of composable morphisms f, g.

Equivalently, a contravariant functor $F : \mathcal{C} \to \mathcal{D}$ is an ordinary covariant functor

$$F : \mathcal{C}^{\text{op}} \to \mathcal{D}. \tag{22}$$

Example 1.9 (Covariant and Contravariant Functors)

The following are examples.

1. The forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$ is covariant, since a homomorphism $f : G \to H$ is mapped to a function $U(f) : U(G) \to U(H)$.
2. The dual space construction is a contravariant functor

$$(-)^* : \mathbf{Vect}_k \to \mathbf{Vect}_k \,. \tag{23}$$

It maps every vector space V to its dual

$$V^* = \text{Hom}_k(V, k) \tag{24}$$

and every linear transformation $T : V \to W$ to its dual transformation

$$T^* : W^* \to V^*, \qquad \varphi \mapsto \varphi \circ T. \tag{25}$$

This reverses the direction of morphisms, as depicted by

$$\begin{aligned} V &\xrightarrow{\quad T \quad} W \\ V^* &\xleftarrow{\quad T^* \quad} W^* \end{aligned} \tag{26}$$

Moreover,

$$(S \circ T)^* = T^* \circ S^*. \tag{27}$$

3. The power set construction defines a contravariant functor

$$\mathcal{P} : \mathbf{Set} \to \mathbf{Set} \tag{28}$$

by associating each set X with its power set $\mathcal{P}(X)$, and each function $f : X \to Y$ with the inverse-image function

$$f^{-1} : \mathcal{P}(Y) \to \mathcal{P}(X). \tag{29}$$

Since

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}, \tag{30}$$

composition is reversed.

Definition 1.13 (Hom-Functors)

Let $\mathcal{C}$ be a locally small category, and fix an object $A \in \text{Ob}(\mathcal{C})$. There are two canonical **Hom-functors**.

1. *Covariant Hom-Functor.* The functor

$$H_A = \text{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \to \mathbf{Set} \tag{31}$$

is defined on objects by

$$H_A(X) = \text{Hom}_{\mathcal{C}}(A, X) \tag{32}$$

and on morphisms $f : X \to Y$ by

$$H_A(f) : \mathrm{Hom}_{\mathcal{C}}(A, X) \to \mathrm{Hom}_{\mathcal{C}}(A, Y), \qquad g \mapsto f \circ g. \tag{33}$$

Thus, $H_A(f)$ acts by postcomposition with f.

$$A \xrightarrow{g} X \xrightarrow{f} Y \qquad (f \circ g : A \to Y) \tag{34}$$

2. *Contravariant Hom-Functor.* The functor

$$H^A = \mathrm{Hom}_{\mathcal{C}}(-, A) : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set} \tag{35}$$

is defined on objects by

$$H^A(X) = \mathrm{Hom}_{\mathcal{C}}(X, A) \tag{36}$$

and on morphisms $f : X \to Y$ in $\mathcal{C}$ by

$$H^A(f) : \mathrm{Hom}_{\mathcal{C}}(Y, A) \to \mathrm{Hom}_{\mathcal{C}}(X, A), \qquad g \mapsto g \circ f. \tag{37}$$

Thus, $H^A(f)$ acts by precomposition with f.

$$X \xrightarrow{f} Y \xrightarrow{g} A \qquad (g \circ f : X \to A) \tag{38}$$

More generally, the **Hom-bifunctor** is

$$\mathrm{Hom}_{\mathcal{C}}(-, -) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}, \tag{39}$$

which is contravariant in its first argument and covariant in its second argument.

Example 1.10 (Hom-Functors)

The following are examples of Hom-functors.

1. In **Set**, fixing a set A gives the covariant Hom-functor

$$H_A(X) = \mathrm{Hom}_{\mathbf{Set}}(A, X). \tag{40}$$

For every function $f : X \to Y$, it maps each function $g : A \to X$ to $f \circ g$.

2. In **Grp**, fixing $A = \mathbb{Z}$ gives

$$H_{\mathbb{Z}}(G) = \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z}, G). \tag{41}$$

Every homomorphism $\varphi : \mathbb{Z} \to G$ is uniquely determined by $\varphi(1)$. Therefore, there is a natural bijection

$$H_{\mathbb{Z}}(G) \cong U(G), \qquad \varphi \mapsto \varphi(1), \tag{42}$$

where $U : \mathbf{Grp} \to \mathbf{Set}$ is the forgetful functor. Hence,

$$H_{\mathbb{Z}} \cong U \tag{43}$$

as functors.

3. In $\mathbf{Vect}_k$, fixing $A = k$ gives the covariant Hom-functor

$$H_k(V) = \mathrm{Hom}_k(k, V). \tag{44}$$

Every linear transformation $T : k \to V$ is uniquely determined by $T(1)$. Therefore,

$$\mathrm{Hom}_k(k, V) \cong V \tag{45}$$

naturally as sets.

4. In $\mathbf{Vect}_k$, the dual space construction is the contravariant Hom-functor

$$(-)^* = \mathrm{Hom}_k(-, k). \tag{46}$$

It sends a linear transformation $T : V \to W$ to the dual map $T^* : W^* \to V^*$.

1.3 Natural Transformations

So far, general categories and functors are pretty abstract, so we would like to represent them in some category we are familiar with, say **Set**. Let $\mathcal{C}$ be a locally small category and choose an object $A \in \mathrm{Ob}(\mathcal{C})$. Now consider the functor

$$H_A : \mathcal{C} \to \mathbf{Set} \tag{47}$$

We must define its actions on objects and morphisms in $\mathcal{C}$.

1. Given an object $X \in \mathrm{Ob}(\mathcal{C})$, we define

$$H_A(X) \coloneqq \mathrm{Hom}_{\mathcal{C}}(A, X) \tag{48}$$

which is viable since $\mathrm{Hom}_{\mathcal{C}}(A, X) = \{g : A \to X\}$ is a set, i.e. an object in **Set**.

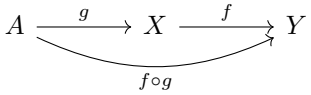
2. Let $f : X \to Y$ be a morphism in $\mathcal{C}$. To define its actions on functions, note that by definition of a functor, we must have $H_A(f) : H_A(X) \to H_A(Y)$, that is

$$H_A(f) : \mathrm{Hom}_{\mathcal{C}}(A, X) \to \mathrm{Hom}_{\mathcal{C}}(A, Y) \tag{49}$$

$H_A(f)$ should be a morphism in **Set**, i.e. a function, and this function will take morphisms $g : A \to X$ and must output a morphism $h : A \to Y$. This should give us a hint on how to define $H_A(f)$. Recall that if we have a morphisms $g : A \to X$ and $f : X \to Y$ in a category, we must have the composition $f \circ g : A \to Y$. Therefore, we can define

$$H_A(f)(g) = f \circ g \tag{50}$$

aka postcomposition of g with f.



Therefore, we have transferred all objects and morphisms in $\mathcal{C}$ to objects and morphisms in **Set**. We can consider

1. objects $X \in \mathrm{Ob}(\mathcal{C})$ as sets $\mathrm{Hom}_{\mathcal{C}}(A, X) = \{g : A \to X\}$.
2. morphisms $f : X \to Y$ as functions $H_A(f) : H_A(X) \to H_A(Y)$.
3. compositions $f \circ g$ as compositions of functions $H_A(f) \circ H_A(g)$.

Definition 1.14 (Functor Category)

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. The **functor category**, denoted $[\mathcal{C}, \mathcal{D}]$, is the category defined by

1. *Objects.* Functors $F : \mathcal{C} \to \mathcal{D}$.
2. *Morphisms.* Natural transformations $\eta : F \Rightarrow G$ between functors.
3. *Composition.* Given natural transformations $\eta : F \Rightarrow G$ and $\theta : G \Rightarrow H$, their composition

$\theta \circ \eta : F \Rightarrow H$ is defined componentwise by

$$(\theta \circ \eta)_X = \theta_X \circ \eta_X. \tag{51}$$

4. *Identity.* For every functor F, the identity natural transformation $1_F : F \Rightarrow F$ is defined by

$$(1_F)_X = 1_{F(X)}. \tag{52}$$

When size conditions are appropriate, these data form a category. In particular, if $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, the functor category is locally small.
The isomorphisms in $[\mathcal{C}, \mathcal{D}]$ are precisely the natural isomorphisms.

Example 1.11 (Functor Categories)

The following are examples of functor categories.

1. The category $[\mathcal{C}, \mathbf{Set}]$ has set-valued functors as objects and natural transformations between them as morphisms.
2. The category $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ has contravariant set-valued functors as objects and natural transformations between them as morphisms. Its objects are called **presheaves** on $\mathcal{C}$.
3. If $\mathcal{C}$ is the category with two objects $0, 1$ and one nonidentity morphism $0 \to 1$, then an object of $[\mathcal{C}, \mathbf{Set}]$ is a function $f : X \to Y$. A morphism between two such objects $f : X \to Y$ and $g : X' \to Y'$ is a pair of functions $a : X \to X'$ and $b : Y \to Y'$ satisfying

$$g \circ a = b \circ f. \tag{53}$$

Equivalently, it is a commutative square

$$\begin{array}{ccc} X & \xrightarrow{a} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ Y & \xrightarrow[b]{} & Y' \end{array} \tag{54}$$

Thus, this functor category is the arrow category of **Set**.

Definition 1.15 (Natural Transformation)

For two functors $F, G : \mathcal{C} \to \mathcal{D}$, a **natural transformation** is a family of morphisms $\eta = (\eta_X)_{X \in \text{Ob}(\mathcal{C})}$ where $\eta_X \in \text{Hom}_{\mathcal{D}}(F(X), G(X))$.[a] It must satisfy the following properties:

1. It assigns every object $X \in \text{Ob}(\mathcal{C})$ to a morphism in $\mathcal{D}$.

$$X \mapsto \eta_X : F(X) \to G(X) \tag{55}$$

2. The assignments above much be such that for any $f : X \to Y$ in $\mathcal{C}$, we have

$$\eta_Y \circ F(f) = G(f) \circ \eta_X \tag{56}$$

depicted through the commutative diagram.

$$X \longrightarrow Y \qquad \rightsquigarrow \qquad \begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array}$$

[a] η is a morphism in the *functor category* $[\mathcal{C}, \mathcal{D}]$, where the objects are functors and morphisms are natural transformations.

Example 1.12 (Grp)

Consider $\mathcal{C} = \mathbf{Grp}$, and let us choose $A = \mathbb{Z}$. Then, we can construct the functor by first defining its action on groups $H_{\mathbb{Z}} : \mathbf{Grp} \to \mathbf{Set}$.

$$H_{\mathbb{Z}}(G) = \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z}, G) = \{\varphi : \mathbb{Z} \to G \colon \varphi \text{ is a group homomorphism}\} \tag{57}$$

But this homomorphism is entirely determined by $\varphi(1)$ since $\varphi(n) = \varphi(1)^n$. Moreoever, every element $g \in G$ determines a unique group homomorphism, so we have a natural bijection (as sets)

$$H_{\mathbb{Z}}(G) \cong G, \quad \varphi_g \cong g \tag{58}$$

where $\varphi_g(1) = g$. Now consider a group homomorphism $f : G \to K$. Then, we can define

$$H_{\mathbb{Z}}(f) : \mathrm{Hom}_{\mathbf{Grp}}(\mathbb{Z}, G) \to \mathrm{Hom}_{\mathbf{Grp}}(G, K), \quad H_{\mathbb{Z}}(f)(\varphi_g) \coloneqq f \circ \varphi_g \tag{59}$$

But note that $H_{\mathbb{Z}}(f)(\varphi_g)(1) = (f \circ \varphi_g)(1) = f(\varphi_g(1)) = f(g)$. Therefore, $H_{\mathbb{Z}}(f)$ behaves exactly like the underlying function f! That is, given $H_{\mathbb{Z}}(f)$,

$$g \mapsto H_{\mathbb{Z}}(f)(g)(1) = f(g) \tag{60}$$

So, groups are mapped to their underlying sets, and group homomorphisms are mapped to functions that—when evaluated at g and then evaluated at 1—gives the action of f on g.

But this may not capture the entire category. For an arbitrary choice of A, the functor $H_A : \mathcal{C} \to \mathbf{Set}$ might lose information.

Example 1.13

If $\mathcal{C} = \mathbf{Grp}$ and A is the trivial group, then $\mathrm{Hom}_{\mathbf{Grp}}(A, G)$ contains exactly 1 homomorphism for every group G. Consequently, H_A sends every group to a singleton set and every group homomorphism to the unique function between singleton sets.

Definition 1.16 (Natural Isomorphism)

Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A natural transformation $\eta : F \Rightarrow G$ is called a **natural isomorphism** if every component

$$\eta_X : F(X) \to G(X) \tag{61}$$

is an isomorphism in $\mathcal{D}$.
Equivalently, there exists a natural transformation $\eta^{-1} : G \Rightarrow F$ satisfying

$$\eta_X^{-1} \circ \eta_X = 1_{F(X)}, \qquad \eta_X \circ \eta_X^{-1} = 1_{G(X)} \tag{62}$$

for every $X \in \mathrm{Ob}(\mathcal{C})$.
In this case, F and G are said to be **naturally isomorphic**, denoted

$$F \cong G. \tag{63}$$

Example 1.14 (Natural Isomorphism)

Consider the double dual functor

$$(-)^{**} : \mathbf{FinVect}_k \to \mathbf{FinVect}_k \,. \tag{64}$$

For every finite-dimensional vector space V, define

$$\eta_V : V \to V^{**}, \qquad \eta_V(v)(\varphi) = \varphi(v) \tag{65}$$

for every $\varphi \in V^*$.
Each η_V is a vector space isomorphism. Moreover, for every linear transformation $T : V \to W$,

$$\eta_W \circ T = T^{**} \circ \eta_V. \tag{66}$$

This is represented by the commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{\eta_V} & V^{**} \\ {\scriptstyle T}\downarrow & & \downarrow{\scriptstyle T^{**}} \\ W & \xrightarrow[\eta_W]{} & W^{**} \end{array} \tag{67}$$

Therefore, the identity functor and double dual functor are naturally isomorphic:

$$1_{\mathbf{FinVect}_k} \cong (-)^{**}. \tag{68}$$

1.4 Universal Properties

A **universal property** characterizes a mathematical object through its relationships with other objects and morphisms, rather than through its internal construction. The central idea is that universal properties can often be expressed as initial or terminal objects in suitably constructed categories.

Definition 1.17 (Initial, Terminal, and Universal Objects)

Let $\mathcal{C}$ be a category.

1. An object $I \in \mathrm{Ob}(\mathcal{C})$ is called an **initial object** if for every object $X \in \mathrm{Ob}(\mathcal{C})$, there exists a unique morphism $f : I \to X$.

$$\begin{array}{ccc} & I & \\ {\scriptstyle !}\swarrow & & \searrow{\scriptstyle !} \\ X & & Y \end{array} \tag{69}$$

2. An object $T \in \mathrm{Ob}(\mathcal{C})$ is called a **terminal object** if for every object $X \in \mathrm{Ob}(\mathcal{C})$, there exists a unique morphism $f : X \to T$.

$$\begin{array}{ccc} X & & Y \\ & {\scriptstyle !}\searrow \quad \swarrow{\scriptstyle !} & \\ & T & \end{array} \tag{70}$$

3. An object $U \in \mathrm{Ob}(\mathcal{C})$ is called a **universal object** if it is an initial or terminal object.

If an initial or terminal object exists in a category $\mathcal{C}$, it is unique up to unique isomorphism.

Proof. Suppose I and I' are both initial objects. By initiality, there exist unique morphisms

$$f : I \to I', \qquad g : I' \to I. \tag{71}$$

Then $g \circ f : I \to I$ is an endomorphism of I. Since I is initial, there exists exactly one such endomorphism, which must be 1_I. Therefore,

$$g \circ f = 1_I. \tag{72}$$

Similarly, $f \circ g = 1_{I'}$, so f and g are inverses. Hence $I \cong I'$. The isomorphism is unique by initiality. The proof for terminal objects is analogous to that of initial objects.

Example 1.15 (Initial and Terminal Objects)

The following are examples of initial and terminal objects.

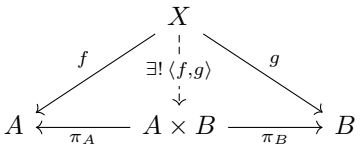
1. In **Set**, the empty set $\varnothing$ is initial, since there exists a unique function $\varnothing \to X$ for every set X. Any singleton set $\{*\}$ is terminal, since there exists a unique function $X \to \{*\}$ for every set X.
2. In **Grp**, the trivial group $\{e\}$ is both initial and terminal, since there exists a unique group homomorphism $\{e\} \to G$ and a unique group homomorphism $G \to \{e\}$ for every group G.
3. In $\mathbf{Vect}_k$, the zero vector space $\{0\}$ is both initial and terminal, since there exists a unique linear transformation from or into $\{0\}$.
4. In **Top**, the empty topological space is initial, while any singleton topological space is terminal.

An object that is both initial and terminal is called a **zero object**.

Suppose a mathematical problem determines a category $\mathcal{D}$ whose objects represent candidate solutions and whose morphisms represent compatible maps between solutions. Thus, to formulate a universal property, we can construct a category of candidate solutions and characterize the desired construction as an initial or terminal object in that category.

Example 1.16 (Cartesian Product as Terminal Universal Object)

Let A, B be sets. For any functions $f : X \to A, g : X \to B$, we know from the universal property of Cartesian products that there exists a unique function $(f, g) : X \to A \times B$ satisfying $f = \pi_A \circ (f, g)$ and $g = \pi_B \circ (f, g)$.



(73)

We wish to restate this in categorical terms, and show their equivalence.

1. ($\to$). Since the result above is true, Consider the category $\mathcal{P}_{A,B}$ consisting of objects $(X, f : X \to A, g : X \to B)$ and morphisms $h : (X, f, g) \to (Y, f', g')$—which are functions $h : X \to Y$—satisfying

   $$f' \circ h = f, \quad g' \circ h = g \tag{74}$$

   We claim that $(A \times B, \pi_A : A \times B \to A, \pi_B : A \times B \to B)$ is a terminal object. Let (X, f, g) be any object in $\mathcal{P}_{A,B}$. Then, we must find a morphism $h : X \to A \times B$ s.t.

   $$\pi_A \circ h = f, \quad \pi_B \circ h = g \tag{75}$$

   By the universal property of the Cartesian product, there exists a unique function $h = (f, g)$ satisfying these equations. Hence, there exists exactly one morphism

   $$(X, f, g) \to (A \times B, \pi_A, \pi_B) \tag{76}$$

in $\mathcal{P}_{A,B}$, proving that $(A \times B, \pi_A, \pi_B)$ is terminal.

2. ($\leftarrow$) now assume that $(A \times B, \pi_A, \pi_B)$ is a terminal object. Then, for any object (X, f, g) there exists a unique morphism $h : (X, f, g) \to (A \times B, \pi_A, \pi_B)$ satisfying

$$\pi_A \circ h = f, \quad \pi_B \circ h = g \tag{77}$$

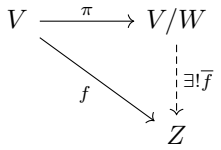
We have proven existence and uniqueness, so we are done. Indeed, we can easily check that the function $h(x) = (f(x), g(x)) = (f, g)(x)$ works, and it must be unique. Therefore, for functions $f : X \to A, g : X \to B$, there exists a unique $h = (f, g) : X \to A \times B$ satisfying

$$\pi_A \circ (f, g) = f, \quad \pi_B \circ (f, g) = g \tag{78}$$

Therefore, we have reformulated our theorem with the equivalent statement that $(A \times B, \pi_A, \pi_B)$ is terminal.

Example 1.17 (Quotient Vector Spaces as Initial Universal Objects)

Let V be a vector space over $\mathbb{F}$, and let W be a subspace. Then, for any linear map $f : V \to Z$ satisfying $W \subset \ker(f)$, by the universal property of quotient spaces, there exists a unique linear map $\overline{f} : V/W \to Z$ satisfying $f = \overline{f} \circ \pi$, where $\pi : V \to V/W$ is the canonical projection.



(79)

We wish to restate this in categorical terms and show their equivalence.

1. ($\rightarrow$). Since the above result is true, consider the category $\mathcal{Q}_W$ consisting of objects $(Z, f : V \to Z)$—where Z is a vector space and f is linear with $W \subset \ker(f)$—and morphisms $h : (Z, f) \to (Z', f')$—which are linear maps $h : Z \to Z'$—satisfying

$$h \circ f = f' \tag{80}$$

We claim that $(V/W, \pi)$ is an initial object in $\mathcal{Q}_W$. Let (Z, f) be any object in $\mathcal{Q}_W$. Then, we must find a morphism $h : V/W \to Z$ such that

$$h \circ \pi = f \tag{81}$$

But we have just proved that $h = \overline{f}$ is the unique map, so $(V/W, \pi)$ must be initial.

2. ($\leftarrow$). Now assume that $(V/W, \pi)$ is an initial object. Then, for any object (Z, f), there exists a unique morphism $h : (V/W, \pi) \to (Z, f)$ satisfying

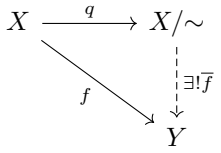
$$h \circ \pi = f \tag{82}$$

Indeed, since $\pi(v) = [v]$, we must have $h([v]) = h(\pi(v)) = f(v)$—which is well-defined since $W \subset \ker(f)$— satisfies this property, so it must be the unique morphism. Therefore, for any linear map $f : V \to Z$ with $W \subset \ker(f)$, there exists a unique linear map $h = \overline{f} : V/W \to Z$ such that $f = \overline{f} \circ \pi$.

Therefore, the universal property of the quotient space is precisely the initiality of $(V/W, \pi)$ in $\mathcal{Q}_W$.

Example 1.18 (Quotient Topological Spaces as Initial Universal Objects)

Let X be a topological space equipped with an equivalence relation $\sim$. Then, from the universal property of quotient spaces, we know that for any topological space Y and continuous map $f : X \to Y$ satisfying $x \sim x' \implies f(x) = f(x')$, there exists a unique continuous map $\overline{f} : X/\sim \to Y$ satisfying $f = \overline{f} \circ q$, where $q : X \to X/\sim$ is the canonical quotient map and $X/\sim$ is equipped with the quotient topology.



(83)

We wish to restate this in categorical terms and show their equivalence.

1. ($\to$). Since the above result is true, consider the category $\mathcal{Q}_\sim$ consisting of objects $(Y, f : X \to Y)$—where Y is a topological space and f is continuous with $x \sim x' \implies f(x) = f(x')$—and morphisms $h : (Y, f) \to (Y', f')$—which are continuous maps $h : Y \to Y'$—satisfying

$$h \circ f = f'. \tag{84}$$

   We claim that $(X/\sim, q)$ is an initial object in $\mathcal{Q}_\sim$. Let (Y, f) be any object in $\mathcal{Q}_\sim$. Then, we must find a morphism $h : X/\sim \to Y$ such that

$$h \circ q = f. \tag{85}$$

   By the universal property of the quotient topological space, there exists a unique continuous map $h = \overline{f}$ satisfying this equation. Hence, there exists exactly one morphism

$$(X/\sim, q) \to (Y, f) \tag{86}$$

   in $\mathcal{Q}_\sim$, proving that $(X/\sim, q)$ is initial.
2. ($\leftarrow$). Now assume that $(X/\sim, q)$ is an initial object. Then, for any object (Y, f), there exists a unique morphism $h : (X/\sim, q) \to (Y, f)$ satisfying

$$h \circ q = f. \tag{87}$$

   We have proven existence and uniqueness, so we are done. Indeed, since $q(x) = [x]$, we must have

$$h([x]) = h(q(x)) = f(x). \tag{88}$$

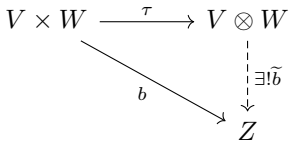
   Therefore, for any continuous map $f : X \to Y$ constant on equivalence classes, there exists a unique continuous map $h = \overline{f} : X/\sim \to Y$ satisfying

$$f = \overline{f} \circ q. \tag{89}$$

Therefore, the universal property of the quotient topological space is precisely the initiality of $(X/\sim, q)$ in $\mathcal{Q}_\sim$.

Example 1.19 (Tensor Products as Initial Universal Objects)

Let V, W be vector spaces over $\mathbb{F}$. We know from the universal property of tensor products that for any vector space Z and bilinear map $b : V \times W \to Z$, there exists a unique linear map $\widetilde{b} : V \otimes W \to Z$ satisfying $b = \widetilde{b} \circ \tau$, where $\tau : V \times W \to V \otimes W$ is the canonical bilinear map defined as the linear extension of $\tau(v, w) = v \otimes w$.



(90)

We wish to restate this in categorical terms and show their equivalence.

1. ($\to$). Since the above result is true, consider the category $\mathcal{T}_{V,W}$ consisting of objects $(Z, b : V \times W \to Z)$—where Z is a vector space and b is bilinear—and morphisms $h : (Z, b) \to (Z', b')$—which are linear maps $h : Z \to Z'$—satisfying

$$h \circ b = b'. \tag{91}$$

   We claim that $(V \otimes W, \tau)$ is an initial object in $\mathcal{T}_{V,W}$. Let (Z, b) be any object in $\mathcal{T}_{V,W}$. Then, we must find a morphism $h : V \otimes W \to Z$ such that

$$h \circ \tau = b. \tag{92}$$

   By the universal property of the tensor product, there exists a unique linear map $h = \widetilde{b}$ satisfying this equation. Hence, there exists exactly one morphism

$$(V \otimes W, \tau) \to (Z, b) \tag{93}$$

   in $\mathcal{T}_{V,W}$, proving that $(V \otimes W, \tau)$ is initial.
2. ($\leftarrow$). Now assume that $(V \otimes W, \tau)$ is an initial object. Then, for any object (Z, b), there exists a unique morphism $h : (V \otimes W, \tau) \to (Z, b)$ satisfying

$$h \circ \tau = b. \tag{94}$$

   We have proven existence and uniqueness, so we are done. Indeed, since $\tau(v, w) = v \otimes w$, we must have

$$h(v \otimes w) = h(\tau(v, w)) = b(v, w). \tag{95}$$

   Therefore, for any bilinear map $b : V \times W \to Z$, there exists a unique linear map $h = \widetilde{b} : V \otimes W \to Z$ satisfying

$$b = \widetilde{b} \circ \tau. \tag{96}$$

Therefore, the universal property of the tensor product is precisely the initiality of $(V \otimes W, \tau)$ in $\mathcal{T}_{V,W}$.

These examples illustrate the same underlying principle: a mathematical construction is characterized by the existence of a unique compatible morphism between candidate solutions.